Induced Subgraphs in Ramsey Graphs

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Abstract

In this dissertation, we investigate the "random-like" features of Ramsey graphs, with an emphasis on the existence of induced subgraphs of many different sizes.

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1 Introduction

Classical Ramsey Theory asks for the minimum number of vertices required to guarantee the existence of a clique or an independent set in a graph. We refer these maxima as *Ramsey numbers*.

Definition 1.1. The *Ramsey number* $R(s,t)$ is the minimum number N such that any graph on N vertices contains a clique of size s or an independent set of size t.

The cornerstone of Ramsey theory due to Erdős and Szekeres gives the following upper bound on $R(s,t)$:

Theorem 1.1 (Erdős–Szekeres [4])**.** *For $s,t \geq 2$,*

$$R(s,t) \leq \binom{s+t-2}{s-1}.$$

Following the fact that $\binom{2t-2}{t-1} < 2^{2t}$, we can see $R(t,t) < 4^t$. In other words, any graph on n vertices contains a clique or independent set of size $\frac{1}{2}\log n$. A central difficulty in Ramsey Theory is the construction for lower bounds for Ramsey numbers. A rather basic result which already gives the best known exponents is the famous random graph construction due to Erdős, marking one of the first application of the probabilistic method in combinatorics. Before stating the result, we first define the *Erdős-Rényi random graph*.

Definition 1.2. For $n \geq 1$ and $p \in [0,1]$, the *random graph* $G_{n,p}$ is the graph on n vertices in which each of the $\binom{n}{2}$ edges is present independently with probability p.

Remark: For constant p, the random graph $G_{n,p}$ is dense in the sense that it has $\Theta(n^2)$ many edges almost surely.

Theorem 1.2 (Erdős [5])**.** *For $n \geq 3$,*

$$\sqrt{2}^t < R(t,t) < 4^t.$$

Proof. The upper bound follows from Theorem 1.1. For lower bound, set $N = \lfloor \sqrt{2}^t \rfloor$ and consider the random graph $G_{N,1/2}$. The expected number of cliques of size t is

$$\binom{N}{t} \cdot 2^{-\binom{t}{2}} < \frac{N^t}{t!} 2^{-\binom{t}{2}} \leq \frac{\sqrt{2}^{t^2}}{t!} 2^{-\binom{t}{2}} = \frac{\sqrt{2}^t}{t!} \leq \frac{\sqrt{8}}{6} < \frac{1}{2}.$$

Similarly, the expected number of independent sets of size t is less than $1/2$. It now follows that there exists a graph on N vertices containing neither a clique nor an independent set of size t. □

In particular, there exists a graph on n vertices containing neither a clique nor an independent set of size $2\log n$, showing that the $\frac{1}{2}\log n$ bound we obtained earlier

could not be substantially improved. The search for explicit graph constructions with logarithmically small cliques and independent sets is a major open problem in Ramsey Theory. maybe cite survey. Despite decades of intensive research, all known constructions still rely on some randomness. This thus motivates the question of whether graphs with small cliques and independent sets inherently resemble dense random graphs like $G_{n,1/2}$; we refer such graphs as *Ramsey graphs*.

Definition 1.3. Let $C > 0$ be a constant. A *C-Ramsey graph* is a graph with no clique or independent set of size greater than $C \log n$.

It is useful to also define a short hand for cliques and independent sets.

Definition 1.4. A set $S \subseteq V(G)$ is *homogeneous* if it induces a clique or an independent set in a graph G.

The first result that cemented the connection between Ramsey graphs and dense random graphs is due to Erdős and Szemerédi, showing that Ramsey graphs are also dense.

Theorem 1.3 (Erdős–Szemerédi [5])**.** *Let $d \in (0, 1/2]$ and let G be a graph on n vertices with less than dn^2 edges. Then G contains a homogeneous set of size greater than $\frac{c}{d \log(1/d)} \log n$ for some constant $c = c(d) > 0$.*

Proof. We may assume that $d < 1/8$, otherwise the Erdős-Szekeres theorem yields the desired homogeneous set for some small enough constant $c > 0$. Note that there exists $m \geq n/2$ vertices in G with degree at most $2dn$. We work over the graph G' induced by these m vertices. Let I be the largest independent set in G' and let $k = |I|$. Note that $\sum_{v \in I} \deg(v) \leq 2dnk \leq 4dmk$. Let $s = 8dk$. Then there are at least $m/2 - k$ vertices $v \in V(G') \setminus I$ with $\deg_I(v) \leq s$. Let J be the set of these vertices. We may assume $s < c_1 \log n / \log(1/d)$, for some sufficiently small c_1, otherwise we are done. Consider the family $\mathcal{F}$ of subsets of I with size less than s. Note that

$$|\mathcal{F}| \leq s \binom{k}{s-1} \leq \frac{s^2}{k} \left(\frac{ek}{s}\right)^s \leq O(\log n) \cdot \left(\frac{e}{8} \cdot \frac{1}{d}\right)^s \leq O(\log n) \cdot \left(\frac{1}{d}\right)^{\frac{c_2 \log n}{\log(1/d)}} \leq n^{1/3},$$

where we pick c_1 small enough such that $c_2 < 1/3$. But then by the pigeonhole principle, there exists a set $Z \subseteq J$ of size at least $(m/2 - k)/n^{1/3} \geq \sqrt{n}$ and some $Y \in \mathcal{F}$ such that $N_I(z) = Y$ for all $z \in Z$. Let $G'' = G[Z]$. Note that G'' does not contain an independent set of size s, otherwise exchange it with Y in I creates a larger independent set in G', contradiction. Recall that $\binom{k}{s-1} \leq \binom{\frac{c_1 \log n}{8d \log(1/d)}}{s-1} \leq n^{1/3} \leq \sqrt{n}$. Thus by Theorem 1.1, G'' contains a clique of size at least

$$\frac{c_1 \log n}{8d \log(1/d)} - s \geq \frac{c \log n}{\log(1/d)},$$

for positive constant $c < (1 - 1/8d)c_1$. $\square$

Corollary 1.4. *For any C-Ramsey graph G, there exists $\delta = \delta(C) \in (0, 1/2)$ such that the edge density of G lies in the interval $[\delta, 1-\delta]$.*

Proof. Apply Theorem 1.3 to G and $\overline{G}$. □

Due to the rich variety of induced subgraphs in dense random graphs, a popular angle to study the pseudorandom properties of Ramsey graphs is to consider their induced subgraphs. Some notable results are a theorem of Shelah [12] which asserts that every n-vertex Ramsey graph contains $2^{\Omega(n)}$ non-isomorphic induced subgraphs, and a theorem of Prömel and Rödl [11] which shows the existence of $\delta = \delta(C) > 0$ such that every C-Ramsey graph on n vertices contains an induced copy of all graphs on at most $\delta \log n$ vertices.

It is also interesting to look at the edge statistics of Ramsey graphs. For a graph G, let

$$\Phi(G) = \{e(H) : H \text{ is an induced subgraph of } G\}$$

Erdős and Mckay [6] conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G).$$

It was proven for Random graphs by Calkin, Frieze, and Mckay [2], but the conjecture for Ramsey graphs remained open for decades and only started to show significant progress in recent years. In particular, a chain of results were sparked after the work of Narayanan, Sahasrabudhe, and Tomon [10] establishing that $|\Phi(G)| = n^{2-o(1)}$ for Ramsey graph G on n vertices. Their proof builds on some ideas from the work of Bukh and Sudakov [1] which showed that every Ramsey graph contains a linear sized subgraph with many distinct degrees, a feature that dense random graphs also exhibit. Kwan and Sudakov [8] later improved the bound to $|\Phi(G)| = \Omega(n^2)$ by refining the argument of Narayanan, Sahasrabudhe, and Tomon. Eventually, Kwan, Sah, Sauermann, and Sawhney [9] closed the conjecture with a fairly involved approach using tools from Fourier analysis, random matrix theory, the theory of Boolean functions.

This dissertation will start by studying the result of Bukh and Sudakov [1] on large induced subgraphs in Ramsey graphs with many distinct degrees. This work not only tightens the connection between Ramsey graphs and random graphs, but also introduces a key feature of Ramsey graphs that unlocks the subsequent $|\Phi(G)| = \Omega(n^2)$ result by Kwan and Sudakov, which we will present in the following chapter. Lastly, we go through the broad strokes of the proof to the Erdős-Mckay conjecture by Kwan, Sah, Sauermann, and Sawhney [9], touching on the core ideas of their approach.

1.1 Notations and Tools

Proposition 1.5 (Turán's theorem)**.** *Every graph G with n vertices and m edges contains an independent set of size at least $n^2/(2m+n)$.*

Proposition 1.6 (Hoeffding's inequality)**.** *Let $X_1, X_2, \ldots, X_n$ be independent real-valued random variables with $0 \leq X_i \leq 1$ for each i, and write $X = \sum_{i=1}^n X_i$. Then for $t \geq 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq 2e^{-2t^2/n}.$$

Proposition 1.7. *Let X denote the number of successes in D draws, without replacement, from a population of size N which contains M successes. Then $\mathbb{E}[X] = DM/N$ and for $t \geq 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq 2e^{-2t^2 D}.$$

Proof. □

2 Subgraphs with Many Degrees

In this chapter, we present the following result of Bukh and Sudakov.

Theorem 2.1 (Bukh–Sudakov [1])**.** *Every Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees.*

This was originally conjectured by Erdős, Faudree, and Sós [6] in light of the observation that the random graph $G_{n,1/2}$ exhibits such a property almost surely, but the $\Omega(\sqrt{n})$ bound is in fact not optimal for both Ramsey graphs and $G_{n,1/2}$. In particular, an upper bound of $O(n^{2/3})$ is noted in [1], which was later proved to be the correct order for Ramsey graphs by Jenssen, Keevash, Long, and Yepremyan [7] via a continuous relaxation approach. The corresponding order for $G_{n,1/2}$ was also matched in an unpublished work of Conlon, Morris, Samotij, and Saxton [3], further reinforcing the case that Ramsey graphs resemble random graphs in various respects. We will, however, focus on the original result of Bukh and Sudakov, as it introduces a key characteristic of Ramsey graphs that will be insightful in the following chapter.

Definition 2.1. An n-vertex graph G is (c, δ)-*diverse* if for $x \in V(G)$, we have $|N(x)\Delta N(y)| < cn$ for at most n^δ vertices $y \in V(G)$.

The notion of diversity captures the idea that the neighbourhoods of most pairs of vertices differ significantly. In particular, the usefulness of diversity comes from bounding the number of vertices that shares the same degree into some random vertex set U, as most pairs of vertices would have fairly uncorrelated neighbourhoods into U.

The proof of Theorem 2.1 proceeds by showing that every Ramsey graph contains a diverse subgraph of linear size, followed by a probabilistic argument that diversity furnishes many distinct degrees.

Lemma 2.2. *For $\delta \in (0, 1)$, there exists $c = c(C, \delta) > 0$ such that every C-Ramsey graph G on n vertices contains a (c, δ)-diverse subgraph on $\Omega(n)$ vertices.*

Proof. Suppose not. Let $K = \lceil \gamma C \log(C + 1) \rceil$ for some sufficiently large constant $\gamma = \gamma(\delta)$, $m = 8^{-K}n$, and $c = 1/K$. We inductively construct a sequence of disjoint vertex sets $S_1, S_2, \ldots, S_K$ of size m^δ, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 8c$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 8c$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G_0 \subset G_1 \subset \cdots \subset G_K$, such that $|G_i| \geq |G_{i-1}|/8$ and $S_i \subseteq V(G_{i-1})$ for all i.

Suppose $G_1, \ldots, G_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. By induction, $|G_{i-1}| \geq 8^{1-i}|G| \geq m$. Note that G_{i-1} is not (c, δ)-diverse by assumption, so there exist vertex $x \in V(G_{i-1})$ and set $S_i \subseteq G_{i-1}$ of size m^δ such that $|N_{G_{i-1}}(x)\Delta N_{G_{i-1}}(y)| < c|G_{i-1}|$

for $y \in S_i$. Let $T_i = V(G_{i-1}) \setminus S_i$. Since $m^\delta \le m/2$ for large enough n, we have $|T_i| \ge |G_{i-1}|/2$.

Suppose $|\overline{N}_{T_i}(x)| \ge |T_i|/2$. Then for $y \in S_i$,

$$d(y, \overline{N}_{T_i}(x)) = \frac{|N_{T_i}(y) \cap \overline{N}_{T_i}(x)|}{|N_{T_i}(x)|} < \frac{c|G_{i-1}|}{|T_i|/2} \le 4c.$$

Partition $\overline{N}_{T_i}(x)$ into $V_i \sqcup W_i$, where $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) < 8c\}$. Notice

$$8c|W_i| \le \sum_{v \in W_i} d(v, S_i) \le \frac{e(\overline{N}_{T_i}(x), S_i)}{|S_i|} = \frac{|\overline{N}_{T_i}(x)|}{|S_i|} \sum_{v \in S_i} d(y, \overline{N}_{T_i}(x)) \le 4c|\overline{N}_{T_i}(x)|,$$

so $|W_i| \le |\overline{N}_{T_i}(x)|/2$. But then $|V_i| \ge |\overline{N}_{T_i}(x)|/2 \ge |T_i|/4 \ge |G_{i-1}|/8$. Set $G_i = G[V_i]$. Since $d(S_i, U) < 8c$ for all $U \subseteq V_i$, condition (i) holds for S_i. In the case that $|N_{T_i}(x)| < |T_i|/2$, the same argument applies by setting $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) > 1 - 8c\}$, which yields $|G_i| \ge |G_{i-1}|/8$ and S_i satisfying condition (ii). This completes the induction.

By working on $\overline{G}$ instead of G if necessary, we may assume that $r \ge K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^r S_{i_k}$. Since each S_{i_k} has size m^δ and $c = 1/K$,

$$d(S) \le \frac{\binom{m^\delta}{2}}{\binom{rm^\delta}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \le \frac{1}{r} + 8c \le \frac{10}{K}.$$

Thus Theorem 1.3 yields a homogeneous set A in S with $|A| \ge \frac{\nu K}{10 \log(K/10)} \log |S|$, for some $\nu = \nu(C)$. Note that $|A|$ is also a homogeneous set in G. But then

$$\log |S| \ge \frac{1}{\delta} \log m = \frac{1}{\delta} \log n - \frac{K \log 8}{\delta}.$$

Pick $\gamma > \nu K/10\delta \log(K/10)$ and we have $|A| > C \log n$ for large enough n, contradiction. □

Lemma 2.3. *Let $\delta \in (0, 1/2)$ and let G be a (c, δ)-diverse graph on n vertices for large enough n. Then for $n/4 \le m \le 3n/4$, there is a subgraph on m vertices containing vertices of $\sqrt{cn}/320$ distinct degrees.*

Proof. Fix m as above. Let U be chosen uniformly at random among all subsets of $V(G)$ of size m. Let $H = G[U]$. It suffices to show that with positive probability, H contains $\Omega(\sqrt{n})$ vertices of distinct degrees.

For distinct $x, y \in V(G)$, define $\mathbb{1}_{xy}$ to be the indicator for the event that $x, y \in V(H)$ and $\deg_H(x) \ne \deg_H(y)$. Then $\mathbb{E}[\mathbb{1}_{xy}] \le \mathbb{P}[\deg_U(x) = \deg_U(y)]$. We may write

$$\mathbb{P}[\deg_U(x) = \deg_U(y)] = \mathbb{P}\left[|(N_G(x) \setminus N_G(y)) \cap U| = |(N_G(y) \setminus N_G(x)) \cap U|\right].$$

Put $D = N_G(x)\Delta N_G(y)$, and $k = |D \cap U|$. Suppose $|D| \geq cn$. Since $\frac{|D|}{4} \leq \frac{m|D|}{n} \leq \frac{3|D|}{4}$, Proposition 1.7 yields

$$\mathbb{P}\left[k < \frac{|D|}{8} \text{ or } k > \frac{7|D|}{8}\right] \leq \mathbb{P}\left[|k - \mathbb{E}[k]| > \frac{|D|}{8}\right] \leq e^{-\Theta(|D|)} = e^{-\Theta(n)} = o(n^{-1/2}).$$

But then

$$\mathbb{P}\left[\deg_U(x) = \deg_U(y) \,\middle|\, \frac{|D|}{8} \leq k \leq \frac{7|D|}{8}\right] = \frac{\binom{|N_G(x)\setminus N_G(y)|}{k/2}\binom{|N_G(y)\setminus N_G(x)|}{k/2}}{\binom{|D|}{k}}.$$

It now follows from Stirling's formula and $k(|D| - k) \geq \frac{7}{64}|D|^2$ that the above probabilitydoes not exceed

$$\frac{\binom{|D|/2}{k/2}^2}{\binom{|D|}{k}} = (1 + o(1))\sqrt{\frac{2|D|}{\pi k(|D| - k)}} \leq \frac{3}{\sqrt{|D|}} \leq \frac{3}{\sqrt{cn}}.$$

Thus if $|N_G(x)\Delta N_G(y)| \geq cn$ then $\mathbb{E}[\mathbb{1}_{xy}] \leq 3/\sqrt{cn} + o(n^{-1/2}) \leq 4/\sqrt{cn}$.

Let $I = \sum_{x,y \in V(G)} \mathbb{1}_{xy}$. By definition of a (c, δ)-diverse graph, we have $\mathbb{E}[\mathbb{1}_{xy}] \leq 4/\sqrt{cn}$ for all but at most $n^{1+\delta} < n^{3/2}$ pairs $x, y \in V(G)$. Thus,

$$\mathbb{E}[I] \leq n^{1+\delta} + \frac{4}{\sqrt{cn}} \cdot n^2 \leq \frac{5}{\sqrt{c}} n^{3/2},$$

and so there is a choice for U such that $I \leq 5n^{3/2}/\sqrt{c}$. Now let r be the number of vertices of distinct degrees in H. Let n_k be the number of vertices in H with degree k and note that $\sum_k n_k = m$. By Jensen's inequality

$$5n^{3/2}/\sqrt{c} \geq I = \sum_k \binom{n_k}{2} \geq r\binom{\frac{1}{r}\sum_k n_k}{2} \geq \frac{m^2}{4r} \geq \frac{n^2}{64r}.$$

Rearranging yields the desired result. □

Theorem 2.1 now follows immediately from Lemmas 2.2 and 2.3.

3 Subgraphs of Quadratically Many Sizes

4 The Erdős–McKay Conjecture

5 Conclusing Remarks

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A Supplementary Proofs